

```
-- =====
-- TASK 1. Generate and insert 200 random rows (airport)
-- =====

SELECT count(*) AS before_count FROM airport;

INSERT INTO airport (airport_name, country, state, city)
SELECT 'Airport ' || i,
       'Country ' || (random()*30)::int,
       'State '   || (random()*50)::int,
       'City '    || (random()*100)::int
FROM generate_series(1, 200) AS i;

SELECT count(*) AS after_count FROM airport;
SELECT * FROM airport ORDER BY airport_id DESC LIMIT 20;
```

Output:

airport_id	airport_name	country	state	city	created_at	updated_at
208	Airport 200	Country 24	State 8	City 51	2026-09-30 04:20:31.481208	2026-09-30 04:20:31.481208
207	Airport 199	Country 22	State 2	City 64	2026-09-30 04:20:31.481208	2026-09-30 04:20:31.481208
206	Airport 198	Country 21	State 19	City 82	2026-09-30 04:20:31.481208	2026-09-30 04:20:31.481208
205	Airport 197	Country 5	State 47	City 8	2026-09-30 04:20:31.481208	2026-09-30 04:20:31.481208
204	Airport 196	Country 18	State 44	City 59	2026-09-30 04:20:31.481208	2026-09-30 04:20:31.481208
203	Airport 195	Country 1	State 48	City 75	2026-09-30 04:20:31.481208	2026-09-30 04:20:31.481208
202	Airport 194	Country 29	State 39	City 13	2026-09-30 04:20:31.481208	2026-09-30 04:20:31.481208
201	Airport 193	Country 23	State 34	City 27	2026-09-30 04:20:31.481208	2026-09-30 04:20:31.481208
200	Airport 192	Country 14	State 4	City 40	2026-09-30 04:20:31.481208	2026-09-30 04:20:31.481208
199	Airport 191	Country 8	State 17	City 55	2026-09-30 04:20:31.481208	2026-09-30 04:20:31.481208
198	Airport 190	Country 22	State 46	City 61	2026-09-30 04:20:31.481208	2026-09-30 04:20:31.481208
197	Airport 189	Country 20	State 19	City 63	2026-09-30 04:20:31.481208	2026-09-30 04:20:31.481208
196	Airport 188	Country 17	State 8	City 26	2026-09-30 04:20:31.481208	2026-09-30 04:20:31.481208
195	Airport 187	Country 8	State 5	City 86	2026-09-30 04:20:31.481208	2026-09-30 04:20:31.481208
194	Airport 186	Country 25	State 33	City 73	2026-09-30 04:20:31.481208	2026-09-30 04:20:31.481208
193	Airport 185	Country 6	State 4	City 97	2026-09-30 04:20:31.481208	2026-09-30 04:20:31.481208
192	Airport 184	Country 10	State 47	City 17	2026-09-30 04:20:31.481208	2026-09-30 04:20:31.481208
191	Airport 183	Country 4	State 42	City 21	2026-09-30 04:20:31.481208	2026-09-30 04:20:31.481208
190	Airport 182	Country 17	State 39	City 29	2026-09-30 04:20:31.481208	2026-09-30 04:20:31.481208
189	Airport 181	Country 23	State 45	City 90	2026-09-30 04:20:31.481208	2026-09-30 04:20:31.481208

(20 rows)

```
-- =====
-- TASK 2. Add airline "KazAir" (Kazakhstan)
-- =====

INSERT INTO airline (airline_name, airline_country)
VALUES ('KazAir', 'Kazakhstan');

SELECT * FROM airline;
```

INSERT 0 1

airline_id	airline_code	airline_name	airline_country	created_at	updated_at
1	KC	Air Astana	Kazakhstan	2026-09-30 04:20:31.472305	2026-09-30 04:20:31.472305
2	DV	SCAT Airlines	Kazakhstan	2026-09-30 04:20:31.472305	2026-09-30 04:20:31.472305
3		KazAir	Kazakhstan	2026-09-30 04:20:31.482254	2026-09-30 04:20:31.482254

(3 rows)

```
-- =====
-- TASK 3. Update KazAir country to Turkey
-- =====
```

```
UPDATE airline
SET airline_country = 'Turkey'
WHERE airline_name = 'KazAir';
```

```
SELECT * FROM airline;
```

```
UPDATE 1
airline_id | airline_code | airline_name | airline_country | created_at | updated_at
-----+-----+-----+-----+-----+-----
          1 | KC           | Air Astana   | Kazakhstan      | 2026-09-30 04:20:31.472305 | 2026-09-30 04:20:31.472305
          2 | DV           | SCAT Airlines | Kazakhstan      | 2026-09-30 04:20:31.472305 | 2026-09-30 04:20:31.472305
          3 |              | KazAir       | Turkey          | 2026-09-30 04:20:31.482254 | 2026-09-30 04:20:31.482254
(3 rows)
```

```
-- =====
-- TASK 4. Add three airlines at once
-- =====
```

```
INSERT INTO airline (airline_name, airline_country) VALUES
('AirEasy', 'France'),
('FlyHigh', 'Brazil'),
('FlyFly', 'Poland');
```

```
SELECT * FROM airline;
```

```
INSERT 0 3
airline_id | airline_code | airline_name | airline_country | created_at | updated_at
-----+-----+-----+-----+-----+-----
          1 | KC           | Air Astana   | Kazakhstan      | 2026-09-30 04:20:31.472305 | 2026-09-30 04:20:31.472305
          2 | DV           | SCAT Airlines | Kazakhstan      | 2026-09-30 04:20:31.472305 | 2026-09-30 04:20:31.472305
          3 |              | KazAir       | Turkey          | 2026-09-30 04:20:31.482254 | 2026-09-30 04:20:31.482254
          4 |              | AirEasy      | France          | 2026-09-30 04:20:31.482981 | 2026-09-30 04:20:31.482981
          5 |              | FlyHigh      | Brazil          | 2026-09-30 04:20:31.482981 | 2026-09-30 04:20:31.482981
          6 |              | FlyFly       | Poland          | 2026-09-30 04:20:31.482981 | 2026-09-30 04:20:31.482981
(6 rows)
```

```
-- =====
-- TASK 5. Delete all flights whose arrival is in 2024
-- =====
```

```
SELECT flight_id, sch_arrival_time FROM flights; -- before
```

```
DELETE FROM flights
WHERE EXTRACT(YEAR FROM sch_arrival_time) = 2024;
```

```
SELECT flight_id, sch_arrival_time FROM flights; -- after
```

flight_id	sch_arrival_time
1	2026-10-01 10:00:00
2	2026-10-02 14:00:00
3	2026-10-03 11:00:00
4	2024-05-01 10:00:00

(4 rows)

DELETE 1

flight_id	sch_arrival_time
1	2026-10-01 10:00:00
2	2026-10-02 14:00:00
3	2026-10-03 11:00:00

(3 rows)

```
-- =====
-- TASK 6. Increase all ticket prices by 15%
-- =====
```

```
SELECT booking_id, ticket_price FROM booking ORDER BY booking_id; -- before
```

```
UPDATE booking
```

```
SET ticket_price = ticket_price * 1.15;
```

```
SELECT booking_id, ticket_price FROM booking ORDER BY booking_id; -- after
```

booking_id	ticket_price
101	25000.00
102	25000.00
103	27000.00
104	27000.00
105	18000.00
106	5000.00

(6 rows)

UPDATE 6

booking_id	ticket_price
101	28750.00
102	28750.00
103	31050.00
104	31050.00
105	20700.00
106	5750.00

(6 rows)

```
-- =====
-- TASK 7. Delete all tickets where price < 10000
-- =====
SELECT booking_id, ticket_price FROM booking WHERE ticket_price < 10000;  --
before

DELETE FROM booking
WHERE ticket_price < 10000;

SELECT booking_id, ticket_price FROM booking ORDER BY booking_id;      --
after
```

booking_id	ticket_price
106	5750.00

(1 row)

DELETE 1

booking_id	ticket_price
101	28750.00
102	28750.00
103	31050.00
104	31050.00
105	20700.00

(5 rows)

```
-- =====
-- TASK 8. Default code 'UNK' where airline_code is NULL
-- =====
SELECT airline_id, airline_code, airline_name FROM airline ORDER BY
airline_id;  -- before

UPDATE airline
SET airline_code = 'UNK'
WHERE airline_code IS NULL;

SELECT airline_id, airline_code, airline_name FROM airline ORDER BY
airline_id;  -- after
```

airline_id	airline_code	airline_name
1	KC	Air Astana
2	DV	SCAT Airlines
3		KazAir
4		AirEasy
5		FlyHigh
6		FlyFly

(6 rows)

UPDATE 4

airline_id	airline_code	airline_name
1	KC	Air Astana
2	DV	SCAT Airlines
3	UNK	KazAir
4	UNK	AirEasy
5	UNK	FlyHigh
6	UNK	FlyFly

(6 rows)

-- TASK 9. Delete baggage checks before 2023-06-01 with 'Not checked'

-- =====

SELECT * FROM baggage_check ORDER BY baggage_check_id; -- before

```
DELETE FROM baggage_check
WHERE created_at < '2023-06-01'
AND check_result = 'Not checked';
```

SELECT * FROM baggage_check ORDER BY baggage_check_id; -- after

baggage_check_id	check_result	created_at	updated_at	booking_id	passenger_id
1	Passed	2026-09-30 04:20:31.475351	2026-09-30 04:20:31.475351	101	1
2	Passed	2026-09-30 04:20:31.475351	2026-09-30 04:20:31.475351	102	2
3	Passed	2026-09-30 04:20:31.475351	2026-09-30 04:20:31.475351	103	3
4	Passed	2026-09-30 04:20:31.475351	2026-09-30 04:20:31.475351	104	4
5	Passed	2026-09-30 04:20:31.475351	2026-09-30 04:20:31.475351	105	5
6	Not checked	2023-05-10 12:00:00	2026-09-30 04:20:31.480501	101	1
7	Not checked	2023-01-15 09:30:00	2026-09-30 04:20:31.480501	102	2
8	Not checked	2024-03-15 10:00:00	2026-09-30 04:20:31.480501	103	3
9	Not checked	2024-03-20 16:45:00	2026-09-30 04:20:31.480501	104	4

(9 rows)

DELETE 2

baggage_check_id	check_result	created_at	updated_at	booking_id	passenger_id
1	Passed	2026-09-30 04:20:31.475351	2026-09-30 04:20:31.475351	101	1
2	Passed	2026-09-30 04:20:31.475351	2026-09-30 04:20:31.475351	102	2
3	Passed	2026-09-30 04:20:31.475351	2026-09-30 04:20:31.475351	103	3
4	Passed	2026-09-30 04:20:31.475351	2026-09-30 04:20:31.475351	104	4
5	Passed	2026-09-30 04:20:31.475351	2026-09-30 04:20:31.475351	105	5
8	Not checked	2024-03-15 10:00:00	2026-09-30 04:20:31.480501	103	3
9	Not checked	2024-03-20 16:45:00	2026-09-30 04:20:31.480501	104	4

(7 rows)

-- TASK 10. Delete airports: state IS NULL and city Mlawe / Kepuh

-- =====

SELECT * FROM airport WHERE city IN ('Mlawe', 'Kepuh'); -- before

```
DELETE FROM airport
WHERE state IS NULL
AND city IN ('Mlawe', 'Kepuh');
```

SELECT * FROM airport WHERE city IN ('Mlawe', 'Kepuh'); -- after

airport_id	airport_name	country	state	city	created_at	updated_at
4	Mlawe Airport	Poland		Mlawe	2026-09-30 04:20:31.480813	2026-09-30 04:20:31.480813
5	Kepuh Airport	Indonesia		Kepuh	2026-09-30 04:20:31.480813	2026-09-30 04:20:31.480813
6	Mlawe Airport 2	Poland	Mazovia	Mlawe	2026-09-30 04:20:31.480813	2026-09-30 04:20:31.480813

(3 rows)

DELETE 2

airport_id	airport_name	country	state	city	created_at	updated_at
6	Mlawe Airport 2	Poland	Mazovia	Mlawe	2026-09-30 04:20:31.480813	2026-09-30 04:20:31.480813

(1 row)

-- TASK 11. Insert unchecked baggage record and return id + created_at

-- =====

```
INSERT INTO baggage_check (check_result, booking_id, passenger_id)
VALUES ('Not checked', 105, 5)
RETURNING baggage_check_id, created_at;
```

SELECT * FROM baggage_check ORDER BY baggage_check_id DESC LIMIT 5;

```

baggage_check_id |      created_at
-----+-----
10 | 2026-09-30 04:20:31.485983
(1 row)

```

INSERT 0 1

```

baggage_check_id | check_result |      created_at      |      updated_at      | booking_id | passenger_id
-----+-----+-----+-----+-----+-----
10 | Not checked | 2026-09-30 04:20:31.485983 | 2026-09-30 04:20:31.485983 | 105 | 5
9 | Not checked | 2024-03-20 16:45:00 | 2026-09-30 04:20:31.480501 | 104 | 4
8 | Not checked | 2024-03-15 10:00:00 | 2026-09-30 04:20:31.480501 | 103 | 3
5 | Passed | 2026-09-30 04:20:31.475351 | 2026-09-30 04:20:31.475351 | 105 | 5
4 | Passed | 2026-09-30 04:20:31.475351 | 2026-09-30 04:20:31.475351 | 104 | 4
(5 rows)

```

-- TASK 12. Convert all airline countries to uppercase

-- =====

```

SELECT airline_id, airline_name, airline_country FROM airline ORDER BY
airline_id; -- before

```

UPDATE airline

SET airline_country = UPPER(airline_country);

```

SELECT airline_id, airline_name, airline_country FROM airline ORDER BY
airline_id; -- after

```

```

airline_id | airline_name | airline_country
-----+-----+-----
1 | Air Astana | Kazakhstan
2 | SCAT Airlines | Kazakhstan
3 | KazAir | Turkey
4 | AirEasy | France
5 | FlyHigh | Brazil
6 | FlyFly | Poland
(6 rows)

```

UPDATE 6

```

airline_id | airline_name | airline_country
-----+-----+-----
1 | Air Astana | KAZAKHSTAN
2 | SCAT Airlines | KAZAKHSTAN
3 | KazAir | TURKEY
4 | AirEasy | FRANCE
5 | FlyHigh | BRAZIL
6 | FlyFly | POLAND
(6 rows)

```

-- TASK 13. Update airline_id = 5

-- =====

```

SELECT * FROM airline WHERE airline_id = 5; -- before

```

```
UPDATE airline
SET airline_name = 'Global Airways',
    airline_country = 'United Kingdom',
    updated_at = CURRENT_DATE
WHERE airline_id = 5;
```

```
SELECT * FROM airline WHERE airline_id = 5; -- after
```

airline_id	airline_code	airline_name	airline_country	created_at	updated_at
5	UNK	FlyHigh	BRAZIL	2026-09-30 04:20:31.482981	2026-09-30 04:20:31.482981

(1 row)

```
UPDATE 1
```

airline_id	airline_code	airline_name	airline_country	created_at	updated_at
5	UNK	Global Airways	United Kingdom	2026-09-30 04:20:31.482981	2026-09-30 00:00:00

(1 row)

```
-- TASK 14. Set state = 'Capital District' for Astana, London, Tokyo
```

```
-- =====
```

```
SELECT * FROM airport WHERE city IN ('Astana', 'London', 'Tokyo'); -- before
```

```
UPDATE airport
SET state = 'Capital District'
WHERE city IN ('Astana', 'London', 'Tokyo');
```

```
SELECT * FROM airport WHERE city IN ('Astana', 'London', 'Tokyo'); -- after
```

airport_id	airport_name	country	state	city	created_at	updated_at
2	Astana Airport	Kazakhstan	Astana	Astana	2026-09-30 04:20:31.472632	2026-09-30 04:20:31.472632
7	Heathrow	UK	Greater London	London	2026-09-30 04:20:31.480813	2026-09-30 04:20:31.480813
8	Haneda	Japan	Kanto	Tokyo	2026-09-30 04:20:31.480813	2026-09-30 04:20:31.480813

(3 rows)

```
UPDATE 3
```

airport_id	airport_name	country	state	city	created_at	updated_at
2	Astana Airport	Kazakhstan	Capital District	Astana	2026-09-30 04:20:31.472632	2026-09-30 04:20:31.472632
7	Heathrow	UK	Capital District	London	2026-09-30 04:20:31.480813	2026-09-30 04:20:31.480813
8	Haneda	Japan	Capital District	Tokyo	2026-09-30 04:20:31.480813	2026-09-30 04:20:31.480813

(3 rows)

```
-- TASK 15. March 2024 'Not checked' -> 'Checked'
```

```
-- =====
```

```
SELECT * FROM baggage_check
WHERE created_at >= '2024-03-01' AND created_at < '2024-04-01'; -- before
```

```
UPDATE baggage_check
SET check_result = 'Checked'
WHERE created_at >= '2024-03-01' AND created_at < '2024-04-01'
    AND check_result = 'Not checked';
```

```
SELECT * FROM baggage_check
WHERE created_at >= '2024-03-01' AND created_at < '2024-04-01';  -- after
```

baggage_check_id	check_result	created_at	updated_at	booking_id	passenger_id
8	Not checked	2024-03-15 10:00:00	2026-09-30 04:20:31.480501	103	3
9	Not checked	2024-03-20 16:45:00	2026-09-30 04:20:31.480501	104	4

(2 rows)

UPDATE 2

baggage_check_id	check_result	created_at	updated_at	booking_id	passenger_id
8	Checked	2024-03-15 10:00:00	2026-09-30 04:20:31.480501	103	3
9	Checked	2024-03-20 16:45:00	2026-09-30 04:20:31.480501	104	4

(2 rows)